ENSIA/Artificial Intelligence and Data Science/Semester 7
Introduction to Mobile Robotics/ Dr. F. Yacef & Dr. R. Benrezki
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Tutorial 1: Homogenous transformations

Exercise 1

1) The vector $\overrightarrow{OP}$ with coordinates $[0, 1, 0]^T$ undergoes successively a rotation of 90° around the x-axis and 90° around the y-axis. Give the global rotation matrix.

   Check graphically.

2) Find the components of the vector $\overrightarrow{OP} = [1, 1, 0]^T$ after a translation of $[0, 0, 1]^T$ followed by a rotation of 60° around the z-axis.

Exercise 2

1) Determine the transformation matrix A corresponding to a rotation around the x-axis by an angle $\theta = 30°$, then a translation along the y-axis by a length $d = 3\,\text{m}$.

2) Determine the transformation matrix A' corresponding to a translation along the y-axis by a length $d = 3\,\text{m}$ followed by a rotation around the x-axis by $\theta = 30°$.

3) Verify graphically that the matrix product is not commutative.

Exercise 3

We make a rotation of $\pi/2$ around the y axis, followed by a translation of $d = 2\,\text{m}$ along the x axis and a rotation of $-\pi/2$ around the z axis.

1) What are the coordinates of the point in the initial (reference) frame knowing that its (homogeneous) coordinates in the final frame are $[0, 3, 0, 1]^T$? Check the result graphically.

2) Knowing the (homogeneous) coordinates of a point $[1, 2, 0, 1]^T$ in the reference frame, what are its coordinates in the final frame? Find the result by two different methods. Check the result graphically.

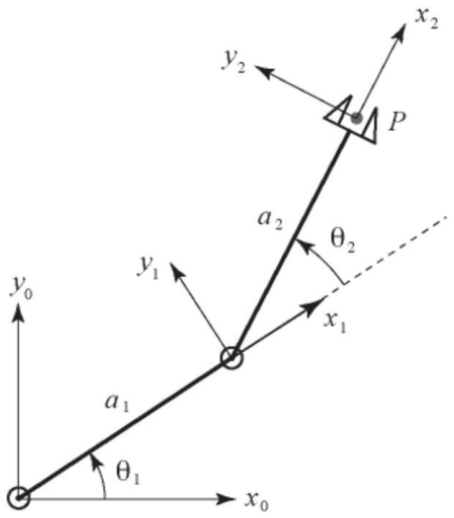


Figure 1. 2 DOF planar robot

Exercise 4

Consider the 2 DOF (RR) planar robot of Figure 1 with a reference frame associated with each segment. Using the homogeneous transformation matrices, determine the position and orientation of the effector organ (point P) relative to the fixed base reference frame (i.e., relative to the O_0, x_0, y_0, z_0 frame).

Exercise 5

Consider a differential drive robot equipped with an onboard sensor (Lidar). We define the following coordinate frames:

- World Frame (W): The global coordinate system where the robot operates.
- Robot Frame (R): The coordinate system attached to the robot, with the origin at the robot's center.
- Sensor Frame (S): The coordinate system attached to the sensor (Lidar), with the origin at the sensor's position.

If the robot's position and orientation in the world frame are given as: $[x_r^W, y_r^W, \theta_r^W] = [3, 2, 45°]$, and the Lidar is mounted 0.25 meter in front of the robot along the robot's x-axis.

1) Calculate the homogeneous transformation matrix from the robot frame to the world frame T_{WR}
2) Calculate the homogeneous transformation matrix from the sensor frame to the robot frame T_{RS}
3) Calculate the position and orientation of the sensor in the world frame $[x_s^W, y_s^W, \theta_s^W]$.
4) Consider a point P, detected by the Lidar in the sensor frame, with coordinates: $[x_p^S, y_p^S] = [2, 2]$. Determine the coordinates of point P in the world frame.

Exercise 6

The Pioneer 3-AT mobile robot shown in Figure 2 has two on-board sensors:

A camera, with coordinate system $O_C, x_C y_C z_C$, and a laser, with coordinate system O_L, x_L, y_L, z_L. The coordinate systems $O_W, x_W, y_W z_W$ and O_R, x_R, y_R, z_R denote the world coordinate system and the coordinate system attached to the robot, respectively. For simplicity, we will assume that the origin O_R of the robot coordinate system coincides with the center of gravity of the robot. Knowing that the coordinates of point P in the camera coordinate system are p_C, determine p_L, p_R and p_W.

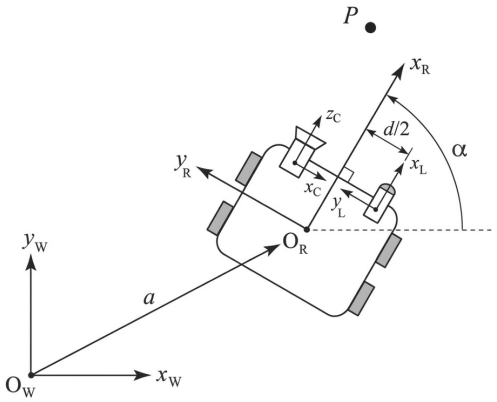


Figure 2. Mobile robot with embedded camera ("C") and laser ("L")